Probabilistic Reasoning Over Time

1 Markov Chains

1.1 Definitions

Stochastic Process A stochastic process is a family of random variables $\{X(t) : t \in T\}$ defined on a *probability space*, where T is the *parameter set*. When $T = \{0, 1, 2, 3, \ldots\}$, it is usually referred to as a time series. When T is a vector space, the stochastic process is called a random field. The collection of all possible states (values) of $X(t)$ forms the *state space*, which can be either discrete or continuous.

Markov Assumption / Markov Property; Markov Process / Markov Chain A Markov process (or Markov chain) is a stochastic process indexed by time, which satisfies the Markov assumption (or Markov property) — namely, the *conditional probability distribution* of the current state depends on only a *finite fixed number* of previous states, i.e.,

$$P(X_t \mid \{X_{t-1}, X_{t-2}, \ldots\}) = P(X_t \mid \{X_{t-1}, X_{t-2}, \ldots, X_{t-k}\}) \quad \text{for any } t$$

First-Order Markov Process A first-order Markov process is a Markov process where the current state depends only on the immediately preceding state, i.e.,

$$P(X_t \mid \{X_{t-1}, X_{t-2}, \ldots\}) = P(X_t \mid X_{t-1}) \quad \text{for any } t$$

In the following sections, we will focus on first-order Markov processes. Without specification, a Markov process is assumed to be first-order.

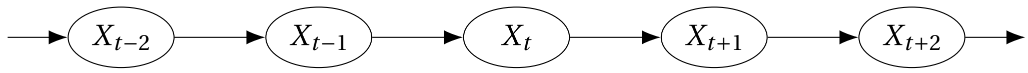

Figure 1: A first-order Markov process.

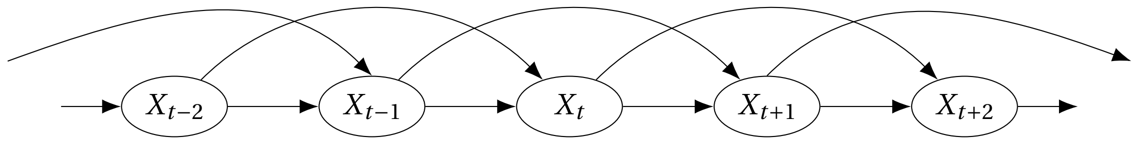

Figure 2: A second-order Markov processs.

Transition Model In a *first-order* Markov process, the transition model is the conditional probability distribution $P(X_t \mid X_{t-1})$. A transition model can be represented in several ways. The following figure shows different representations of the same transition model for a first-order Markov process.

$X_t \backslash X_{t-1}$	Sunny	Rainy
Sunny	0.8	0.3
Rainy	0.2	0.7

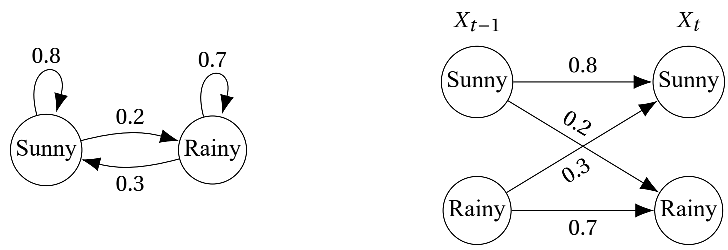

Figure 3: A transition model in three different representations.

From above, we can see that Markov processes are a special class of *Bayesian networks*. And the transition model is the *conditional probability table* (CPT) of the first-order Markov process. Compared to a general Bayesian network, a first-order Markov process has a simpler structure (a chain) and fewer parameters (only one CPT).

1.2 Markov Chain Inference

Inference on Markov chains is basically the same as inference in Bayesian networks.

Forward Algorithm Forward algorithm is a dynamic programming algorithm used to solve queries about the current state given a previous state. It is simply based on the following equation:

$$P(X_t) = \sum_{x \in S} P(X_t \mid X_{t-1} = x) P(X_{t-1} = x) \qquad \text{where} \quad S = \{\text{all possible states}\}$$

Matrix representation The forward algorithm can be expressed in matrix form. Suppose the set of all possible states is given by $S = \{s_1, s_2, \ldots, s_n\}$. Then we can represent the transition model as an $n \times n$ matrix $\boldsymbol{M}$, where the entry $\boldsymbol{M}[i, j] := P(X_t = s_i \mid X_{t-1} = s_j)$. Let the vector $\boldsymbol{p}_t$ represent the probability distribution of X_t, i.e., $\boldsymbol{p}_t[i] := P(X_t = s_i)$. Then the forward algorithm can be expressed as

$$\boldsymbol{p}_t = \boldsymbol{M}\boldsymbol{p}_{t-1} \quad \text{i.e.} \quad \begin{bmatrix} P(X_t = s_1) \\ \vdots \\ P(X_t = s_n) \end{bmatrix} = \begin{bmatrix} P(X_t = s_1 \mid X_{t-1} = s_1) & \cdots & P(X_t = s_1 \mid X_{t-1} = s_n) \\ \vdots & \ddots & \vdots \\ P(X_t = s_n \mid X_{t-1} = s_1) & \cdots & P(X_t = s_n \mid X_{t-1} = s_n) \end{bmatrix} \begin{bmatrix} P(X_{t-1} = s_1) \\ \vdots \\ P(X_{t-1} = s_n) \end{bmatrix}$$

2 Hidden Markov Model (HMM)

2.1 Definitions

When the obseved variables are not the states of the Markov process, but are generated by the states, we have a hidden Markov model (HMM).

Sensor Markov Assumption To make it easier, we assume that the current observation depends only on the current state, i.e.,

$$P(E_t \mid \{X_{t-1}, X_{t-2}, \ldots, E_{t-1}, E_{t-2}, \ldots\}) = P(E_t \mid X_t) \quad \text{for any } t$$

Transition Model & Sensor Model In an HMM, we have two models (two CPTs in the corresponding Bayesian network):

- Transition model (same as in Markov processes): $P(X_t \mid X_{t-1})$
- Sensor model (or observation model): $P(E_t \mid X_t)$

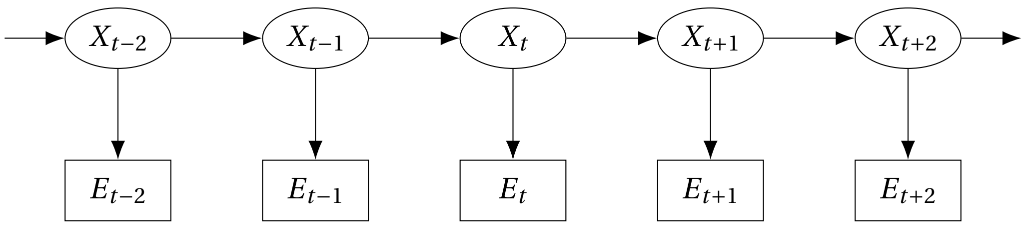

Figure 4: A hidden Markov model (HMM).

2.2 Inference

There are four basic inference tasks in a temporal model:

- **Filtering**: Finding the current state given all evidence up to now, i.e., computing

$$P(X_t \mid E_1 = e_1, E_2 = e_2, \ldots, E_t = e_t) \qquad \text{or simply} \quad P(X_t \mid e_{1:t})$$

- **Prediction**: Finding the future state given all evidence up to now, i.e., computing

$$P(X_{t+k} \mid e_{1:t}) \quad \text{for some } k > 0$$

- **Smoothing**: Finding the past state given all evidence up to now, i.e., computing

$$P(X_k \mid e_{1:t}) \quad \text{for some } 0 \le k < t$$

- **Most likely explanation**: Finding the most likely sequence of states given all evidence up to now, i.e.,

$$\arg\max_{x_{1:t}} P(X_{1:t} = x_{1:t} \mid e_{1:t})$$

In addition to the above inference tasks, we also have **learning**: The transition and sensor models, if not yet known, can be learned from observations.